

Day 5 — Data-Driven Problem Framing

Activity packet for facilitators and participant triads. Day 5 is the integration day: every prior artifact of Week 1 funnels into a problem statement + evidence brief + 10-minute cohort readout.

Prerequisite artifacts (must be brought in)

Each triad arrives with:

- **Prompt Pattern Library** (Day 1)
- **PR/FAQ** (Day 2)
- **Validated Persona Canvas** (Day 3)
- **Scored Pain-Point Matrix + Top 3** (Day 4)
- **Parking lot** of open validation questions (accumulated)

Materials

Problem Statement Template (printable)

Problem statement — [short handle]

****Who**** — [persona role + context]

****Currently**** — [observed behavior, with frequency]

****Which causes**** — [measurable consequence]

****Because**** — [root cause, from 5 Whys]

****We know this because**** — [evidence, tagged by tier]

****Success looks like**** — [the change we'd see if the problem is solved]

Evidence Brief Template

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# Evidence brief – [short handle]

## Persona (one paragraph)
[From Day 3]

## Pain map
[Screenshot or rendering of Day 4 matrix, Top 3 highlighted]

## Problem statement
[From Activity 1]

## Evidence log
| Claim | Source | Date | Tier | Captured by |
|-----|-----|-----|-----|-----|
| ...   | ...   | ...   | ...   | ...   |

## Open questions
- ... (ranked; most impactful first)
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Scoring rubric (used by cohort during readouts)

DIMENSION	WEIGHT	1	3	5
Customer authenticity	25%	Generic	Persona present, mix of claims	Observed-heavy, evidence named
Pain specificity	25%	Categories	Mixed	Verb + circumstance + consequence
Evidence quality	20%	Unsourced	Sources named	Tiered, auditable
Problem statement clarity	20%	Vague or solution-y	Mostly clear	Engineer could scope
AI-use transparency	10%	Unstated	Mentioned	Prompt + validation step disclosed

Activity 1 — Draft Your Statement

Format: Triad • 45 min • Block 1

Purpose

Convert the Week-1 artifact stack (persona + Top 3 pains) into a single six-line problem statement an engineer could scope from.

Setup

Each triad has its Day 3 persona, Day 4 Top 3 pains, and the printed Problem Statement Template.

Steps

1. **Pick your anchor pain (5 min).** Usually #1 from Day 4 Top 3. If you're torn between two, draft both in parallel.
2. **Fill the template (20 min).** One line at a time. Do not skip "Success looks like."
3. **Failure-mode audit (15 min).** Check each line against:
 - Solution in disguise? (If your "Currently" or "Because" implies a specific feature, pull it)
 - Consequence-free? (Add a measurable outcome)
 - No root cause? (Run the 5 Whys again — briefly)
 - Ungrounded? (Tag the evidence)
 - Success-agnostic? (Quantify or time-bound the target)
4. **Within-triad peer read (5 min).** One member reads aloud. Others listen for what's still weak.

Common mistakes to watch for

- **"Because" line restates the problem** — it should name the *mechanism* (why the current state exists at all)
- **"Success looks like" is a feature** — it should be a measurable change in behavior or outcome
- **Persona inflation** — "all dispatchers at all HVAC companies" when your evidence is 3 shops

Deliverable

A six-line problem statement per triad, evidence-tagged, that survives the failure-mode audit.

Facilitation cues

- The "Success looks like" line is the most-skipped. Walk the room and refuse to let it stay empty.
- If a triad has two candidate statements after the parallel draft, ask which one the engineer would prefer to scope from. That usually picks the winner.

Activity 2 — Pressure Test

Format: Triad • 45 min • Block 2

Purpose

Use AI to surface the questions a skeptical engineer would ask — then answer them honestly, including by adding to the Open questions list.

Setup

Each triad has its draft problem statement from Activity 1 and an AI assistant. Provide the pressure-test prompt below.

Steps

1. **Run the prompt (10 min).** Paste in the problem statement and evidence log.
2. **Answer what you can (25 min).** Tag each question green / yellow / red as described below.
3. **Revise (10 min).** Update the problem statement and Evidence log based on answers.

The prompt (from Day 1 library)

Role: Senior engineering lead reading a problem statement I intend to scope against.
Context: [paste problem statement + evidence log]
Constraints:
– Do not rewrite. Ask the questions you would ask me in a scoping meeting.
– Rank your questions by how much they'd change your estimate if unanswered.
– Flag any claim that is inferred or AI-generated that you'd want upgraded first.
Format: Numbered list of questions, ranked.

Protocol

1. **Run the prompt (10 min).**
2. **Answer what you can (25 min).** Go question by question.
 - Green: answered from existing evidence (note the source)
 - Yellow: answerable if we do X (add to Open questions)
 - Red: unknowable from current evidence; flag the assumption we're carrying
3. **Revise (10 min).** Update the problem statement and Evidence log based on answers. Move failed claims to Open questions or drop.

Guardrail

AI does not get to rewrite the statement. Ownership stays with the triad. The prompt's job is to surface weakness, not to patch it.

Deliverable

A revised problem statement plus an updated Evidence log and an expanded Open questions section, with each AI-surfaced question color-tagged.

Facilitation cues

- If a triad has AI rewrite the statement, ask them to point at the line they wrote. Re-anchor ownership.
- A long Open questions list is a feature, not a bug. Honor the triads that grew their list honestly.

Activity 3 — Paired Dry-Run

Format: Paired triads • 45 min • Block 3

Purpose

Rehearse the 10-minute readout against another triad's stopwatch, and surface where intent and listener-summary diverge.

Setup

Pair each triad with another. Each triad needs a stopwatch (phone is fine) and a note-taker on the partner side.

Steps

1. Triads pair up.
2. Triad A delivers a **full timed 10-minute readout** to Triad B. Partner uses a stopwatch.
3. During the readout, Partner writes down:
 - The **3 questions** they would ask in real Q&A

- **One sentence:** "I heard the problem to be..."
- 4. Compare that one-sentence summary to what Triad A intended to convey.
- 5. Swap directions.
- 6. **Joint debrief (15 min):** Where did the heard-summary differ from intent? Why? What slide/paragraph caused the gap?

The "what I heard" test

If the listener's one-sentence summary doesn't match your intent, the readout needs work — usually one of:

- The problem was buried under too much persona setup
- Success criteria were missing or abstract
- The evidence overwhelmed the narrative

Revise accordingly before Block 4.

Deliverable

A polished 10-minute readout plus a documented gap (intent vs. heard-summary) that the triad has addressed before Block 4.

Facilitation cues

- The "what I heard" sentence is the single most diagnostic moment. Slow it down — don't let the listener gloss it.
- If both directions run long, cut the joint debrief to 10 minutes rather than skipping the second readout.

Week 1 Cohort Readouts

Format: Full cohort • ~90 min • Block 4

Structure per triad (10 minutes)

CLOCK	CONTENT
0:00 – 1:30	Persona (one quote, observed behavior)
1:30 – 3:30	Top 3 pains (matrix visual)
3:30 – 5:30	Problem statement (read each line)
5:30 – 7:30	Evidence sample + open questions
7:30 – 10:00	Q&A — questions only , no defending

Between triads (3 min)

Silent. Every audience member fills out a score sheet: rubric scores + 1 strength + 1 suggestion.

Facilitator timing

- 6 triads = ~80 min readouts + 18 min between = **~98 min**. Budget 90 min and stay disciplined.
- Random draw order (pull triad names from a hat or randomizer)
- If running short, compress Q&A to 2 minutes — never the readout itself

Score sheet template

Triad: [name]

Scores (1–5)

- Customer authenticity: __
- Pain specificity: __
- Evidence quality: __
- Problem statement clarity: __
- AI-use transparency: __

One strength

...

One suggestion

...

End-of-Week-1 checkpoint

Each triad leaves the week with:

- ✓ Problem statement + Evidence brief, reviewed by peers and scored
- ✓ Readout deck/visuals reusable on demand
- ✓ A ranked list of Open questions (input to Week 2 research)
- ✓ A clear sense of which evidence tiers need upgrading before Week 3 mini-capstone

Facilitator debrief (end of Friday, instructor only)

Before you log off:

1. Which triads seem **PRD-ready** for Week 3? Which need extra coaching during Week 2?
2. Who was over-relying on AI by the end of the week vs. using it judiciously?
3. Which triad surprised you — in either direction — and why?
4. Three adjustments for Week 2 pacing based on what you saw today.

Bridge to Week 2

Monday opens with: *"Review the problem statements from Friday. Choose your triad's candidate North Star metric. Bring three candidates, come ready to defend one."*